

CLERMONT-FERRAND AUVERGNE AP, France

WMO: 074600

Lat: 45.7869N

Lon: 3.1494E

Elev: 332

StdP: 97.40

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-4.6	-10.0	1.7	-4.1	-7.9	2.0	-2.6	13.4	10.3	11.7	9.8	1.8	350	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	32.8	20.1	30.6	19.6	28.5	19.0	21.3	29.5	20.4	28.5	19.6	27.1	3.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.1	24.3	17.8	13.3	23.1	17.0	12.6	22.5	63.1	29.6	60.1	28.5	57.3	27.3	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	8.8	7.5	-10.5	36.3	2.6	2.0	-12.4	37.7	-13.9	38.8	-15.3	40.0	-17.2	41.4		
			-10.9	23.1	2.5	1.0	-12.7	23.8	-14.2	24.4	-15.6	25.0	-17.5	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	4.2	5.0	8.2	10.7	14.6	18.3	20.4	20.2	16.4	13.1	7.7	4.9
	DBStd	6.98	4.52	4.50	3.89	3.48	3.57	3.83	3.35	3.61	3.45	4.13	4.12	4.45
	HDD10.0	721	183	146	81	32	4	0	0	0	1	18	91	165
	HDD18.3	2544	437	374	315	229	126	48	17	19	78	166	319	417
	CDD10.0	1456	4	5	25	53	146	249	322	316	192	115	22	7
	CDD18.3	237	0	0	0	1	9	47	80	78	19	4	0	0
	CDH23.3	2472	0	0	1	16	123	493	823	799	188	28	0	0
	CDH26.7	891	0	0	0	1	26	173	312	334	43	2	0	0
Wind	WSAvg	2.9	3.1	3.3	3.4	3.2	2.9	2.6	2.4	2.3	2.3	2.9	3.0	3.1
Precipitation	PrecAvg	595	32	27	29	42	89	68	48	74	58	54	38	36
	PrecMax	782	86	65	79	115	152	136	140	149	163	172	84	68
	PrecMin	444	5	4	1	6	18	4	8	12	4	4	11	3
	PrecStd	100	20	14	17	28	35	29	26	36	39	38	21	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	18.3	22.4	25.4	29.7	34.4	35.7	36.1	30.6	26.4	19.7	16.1
		MCWB	11.0	9.7	12.4	15.4	18.1	20.2	20.4	20.2	18.3	16.8	13.5	11.5
	2%	DB	13.9	15.3	19.6	22.6	26.6	31.0	32.4	32.9	27.7	23.7	17.5	14.3
		MCWB	10.1	9.5	11.1	13.8	17.1	19.5	20.2	20.1	18.0	16.4	12.8	10.5
	5%	DB	12.2	13.3	17.1	20.1	24.2	28.4	30.1	30.5	25.5	21.6	15.8	13.0
		MCWB	8.7	8.6	10.5	12.8	16.1	19.3	19.8	19.7	17.1	15.6	11.9	9.7
	10%	DB	10.8	11.7	15.0	17.6	21.9	26.1	27.9	27.9	23.2	19.6	14.0	11.5
		MCWB	7.8	7.8	9.7	11.6	15.1	18.5	18.9	18.9	16.3	14.8	10.9	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.1	13.3	16.3	19.6	22.3	22.4	22.0	20.5	18.2	14.4	12.1
		MCDB	15.1	15.9	20.2	23.7	26.6	29.7	30.5	31.1	26.7	24.0	18.5	15.5
	2%	WB	10.2	9.9	12.0	14.7	18.0	21.1	21.1	21.0	18.9	17.0	13.3	10.8
		MCDB	13.4	14.2	18.0	21.5	24.9	28.8	29.6	29.8	25.2	22.1	16.7	13.9
	5%	WB	9.0	9.0	10.9	13.2	16.7	19.9	20.2	20.2	17.8	16.0	12.2	9.8
		MCDB	12.0	12.9	16.3	18.8	22.7	27.0	28.3	28.6	23.9	20.4	15.2	12.7
	10%	WB	7.8	8.0	9.9	12.0	15.6	18.9	19.4	19.3	16.8	15.1	11.1	8.7
		MCDB	10.5	11.4	14.5	16.8	20.9	24.9	26.8	26.7	22.2	19.1	13.9	11.3
Mean Daily Temperature Range	5% DB	MDBR	7.0	8.4	10.3	10.7	11.1	11.6	12.1	12.1	11.4	9.8	7.9	7.1
		MCDBR	8.9	12.0	14.1	15.4	15.0	15.3	15.9	16.3	14.9	12.4	10.4	8.9
		MCWBR	6.0	7.4	7.9	8.3	7.5	7.0	6.6	6.5	6.9	6.5	6.6	6.1
	5% WB	MCDBR	8.1	10.6	12.4	13.7	13.4	13.8	14.3	14.3	12.8	11.2	9.3	8.3
		MCWBR	5.7	6.9	7.4	7.6	7.2	6.7	6.4	6.3	6.4	6.3	6.2	5.8
Clear-Sky Solar Irradiance	taub	0.295	0.314	0.344	0.373	0.379	0.387	0.382	0.377	0.351	0.345	0.312	0.293	
	taud	2.506	2.453	2.396	2.321	2.349	2.358	2.364	2.389	2.442	2.469	2.512	2.512	
	Ebn at Noon	821	868	885	885	888	880	879	871	867	819	790	784	
	Edh at Noon	66	83	102	120	120	120	118	111	96	82	65	59	
All-Sky Solar Radiation	RadAvg	1.26	2.11	3.30	4.39	5.13	6.01	5.91	5.17	4.01	2.56	1.43	1.08	
	RadStd	0.16	0.30	0.40	0.57	0.50	0.54	0.48	0.43	0.29	0.22	0.16	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page